

The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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June 8, 2026

Abstract

We identify a fundamental structural asymmetry between General Relativity (GR) and Quantum Field Theory (QFT): quantum fields require a fixed kinematic background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field leads to contradictions in background independence and gauge covariance. To resolve this, we propose a minimal, background-independent quantum substrate from which both smooth manifolds and quantum field algebras are conjectured to co-emerge. Geometry is proposed to arise from the entanglement structure of a relational density matrix, and we identify the specific conditions – via known uniqueness and convergence theorems – under which its low-energy collective excitations would take the form of a graviton and its continuum limit would reduce to the Laplace–Beltrami operator and thermodynamic Einstein dynamics. Rather than treating these conditions as automatic, we test both directly on a concrete, minimal realization of the substrate (the spin-1/2 antiferromagnetic Heisenberg chain at its quantum critical point, exact diagonalization, N up to 26), with a gapped chain as a control. The first condition, linear (Lorentz-consistent) low-energy dispersion, is confirmed to high precision ($E_{\text{gap}} \propto N^{-0.96}$, $R^2 = 0.999994$; a quadratic alternative is excluded, $R^2 = 0.009$), matching the exact des Cloiseaux–Pearson result. The second, that the mutual-information matrix encodes a genuine Euclidean metric distance, is falsified for the critical substrate ($R^2 = 0.9998$ for a logarithmic, not linear, fit to the candidate distance) but not featurelessly: the specific functional form obtained is the same one associated with entanglement entropy in 1 + 1D CFT and with hyperbolic, holographically-flavored bulk reconstructions, and it is absent in the gapped control, where the originally hoped-for Euclidean behavior is instead recovered. This framework provides a conceptually coherent, precisely specified, and now partly numerically tested path toward unifying GR and QFT without quantizing the metric as a fundamental field – one open problem resolved in the framework’s favor, the other redirected toward a specific, non-Euclidean alternative rather than closed.

1 Introduction

Quantum Field Theory (QFT) describes matter and radiation as operator-valued distributions defined on a fixed background spacetime [2]. Locality, microcausality, and the construction of a Fock

space all require a predetermined metric structure. General Relativity (GR), by contrast, denies the existence of any prior geometric container: the metric itself is a dynamical field determined by the Einstein equations [3]. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity [1].

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and the very notion of locality. Gauge interactions, by contrast, act only on specific internal symmetry representations. This universality is not merely an empirical observation but a theorem: Weinberg showed that Lorentz invariance and unitarity of the S -matrix alone force any theory containing a massless spin-2 particle to couple to the total energy-momentum tensor with a single coupling constant, the same for every species – the equality of gravitational and inertial mass is a consistency requirement of the S -matrix, not an independent postulate [7]. No analogous universality theorem constrains massless spin-1 particles, which couple only to conserved charges specific to a given internal symmetry and can consistently decouple from any given field. This asymmetry between spin-2 and spin-1 is a piece of genuine, substrate-independent field-theoretic structure, and it is what motivates the Gravitational Baseline Principle: gravity is the universal geometric framework that renders all other interactions definable. It is not a peer field within a background but the background itself.

In this paper, we develop a minimal pre-geometric quantum substrate from which smooth geometry and quantum fields are conjectured to co-emerge. The substrate is defined by a relational density matrix whose entanglement structure encodes emergent spatial connectivity. We identify the precise conditions – via established convergence theorems for graph Laplacians and the Fierz–Pauli uniqueness theorem for massless spin-2 fields – under which the Laplace–Beltrami operator, graviton excitations, and thermodynamic gravitational dynamics would arise in the continuum limit, and we test both directly, via exact diagonalization on a concrete substrate, in Secs. 4.1 and 5: one condition (linear dispersion) is confirmed, the other (Euclidean metric encoding) is falsified in favor of a specific, holographically-motivated alternative.

2 The Structural Asymmetry Problem

Einstein’s field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (1)$$

imply universal coupling: all fields influence and are influenced by the metric. Gauge interactions, however, operate only on specific internal degrees of freedom.

Perturbative quantization of gravity requires expanding the metric as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and quantizing $h_{\mu\nu}$ as a spin-2 gauge field. But this procedure presupposes a fixed background $\eta_{\mu\nu}$, violating diffeomorphism invariance at the foundational level. This is the core structural asymmetry: QFT requires a background; GR denies one.

3 The Gravitational Baseline Principle

Gravity is not a selective interaction but the universal geometric baseline. All physical systems couple non-linearly to the metric, which defines causal structure and measurement. The Planck scales,

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad (2)$$

mark the breakdown of smooth geometry and the onset of pre-geometric structure.

The Gravitational Baseline Principle states:

Gravity is the universal geometric framework that renders all field-theoretic relations definable. It cannot be quantized as a conventional gauge field without violating background independence.

4 Pre-Geometric Substrate Architecture

We define the pre-geometric substrate $\mathcal{Q}$ as a background-independent quantum system with state vector $|\Psi\rangle$ in a Hilbert space $\mathcal{H}_{\mathcal{Q}}$. The density matrix $\rho = |\Psi\rangle\langle\Psi|$ encodes relational structure.

Mutual information between subsystems i and j ,

$$I(i, j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \quad (3)$$

defines a symmetric connectivity matrix A_{ij} .

4.1 Emergent Continuum Geometry

A discrete structural operator acts on test functions $\phi(x)$ mapped to subsystem nodes:

$$\Delta_{\mathcal{Q}} f_i = \sum_j A_{ij} (f_i - f_j). \quad (4)$$

Whether an operator of this form converges to a continuum Laplace–Beltrami operator is not a matter of isotropy alone; it is governed by a precise body of results in spectral graph theory. If the nodes i are points sampled from a Riemannian manifold (M, g) and the weights are given by a kernel of the geodesic distance, $A_{ij} = K_{\epsilon}(d_g(x_i, x_j))$ for a suitable kernel K_{ϵ} (e.g. Gaussian), then as the sampling density grows and the bandwidth $\epsilon \rightarrow 0$ at an appropriate rate, the graph Laplacian provably converges, pointwise and (under further conditions) spectrally, to the weighted Laplace–Beltrami operator of M [10, 11]:

$$\lim_{\epsilon \rightarrow 0} \Delta_{\mathcal{Q}} f = \nabla^2 f + O(\epsilon). \quad (5)$$

This is a rigorous mathematical statement, but its hypotheses are demanding: the edge weights must already be a kernel function of a genuine metric distance on a manifold the nodes are known to sample. In the substrate defined here, A_{ij} is instead fixed directly by the mutual information $I(i, j)$ of Eq. (3), with no manifold or distance function assumed in advance. Eq. (5) therefore does *not* follow automatically from the substrate’s definition; it holds only if some function of $I(i, j)$ – for instance an entanglement-based distance $d_{\mathcal{Q}}(i, j) = -\log(I(i, j)/I_{\max})$, in the spirit of proposals relating entanglement entropy to bulk geodesic distance [6] – can be shown to satisfy, at least asymptotically, the metric and kernel conditions required by the theorems of [10, 11].

Numerical test

We test this directly for a minimal, concrete realization of the substrate: the ground state of the spin-1/2 antiferromagnetic Heisenberg chain, $H = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$ with periodic boundary conditions, at its quantum critical point, using exact diagonalization at $N = 16, 18$. This is the natural first test case: it is gapless (a $c = 1$ conformal field theory in its continuum limit) and has no coordinates or metric assumed anywhere in its definition – $I(i, j)$ is computed purely from the many-body ground state’s entanglement structure. As a control, we repeat the test on the dimerized chain, $H = \sum_i J_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$ with $J_i = 1 + \delta(-1)^i$ at $\delta = 0.3$, whose ground state is gapped and whose correlations decay exponentially rather than algebraically.

On a ring of N sites, the geodesic (chord) distance between sites separated by r hops is $(N/\pi) \sin(\pi r/N)$, not the hop-count r itself; we use this throughout; it is the distance the convergence theorems of [10, 11] actually refer to for this topology. Table 1 reports, for both chains, (i) the goodness of fit of $d_Q(r)$ to a linear function of chord distance (what a Euclidean kernel $A_{ij} \sim e^{-d/\epsilon}$ requires) versus a logarithmic function of chord distance, (ii) the fraction of triples violating the triangle inequality, and (iii) the quality of a classical (Torgerson) embedding of the full distance matrix into 2D Euclidean space.

Table 1: Mutual-information metric-encoding test (Sec. 4.1): candidate distance $d_Q(i, j) = -\log(I(i, j)/I_{\max})$ against chord distance on the ring, exact diagonalization.

N	Chain	linear R^2	log R^2	triangle viol.	MDS-2D R^2
16	critical	0.9471	0.9998	12.4%	0.9313
16	gapped ($\delta = 0.3$)	0.9431	0.9034	6.7%	0.9870
18	critical	0.9411	0.9999	11.0%	0.9246
18	gapped ($\delta = 0.3$)	0.9499	0.8970	5.9%	0.9845

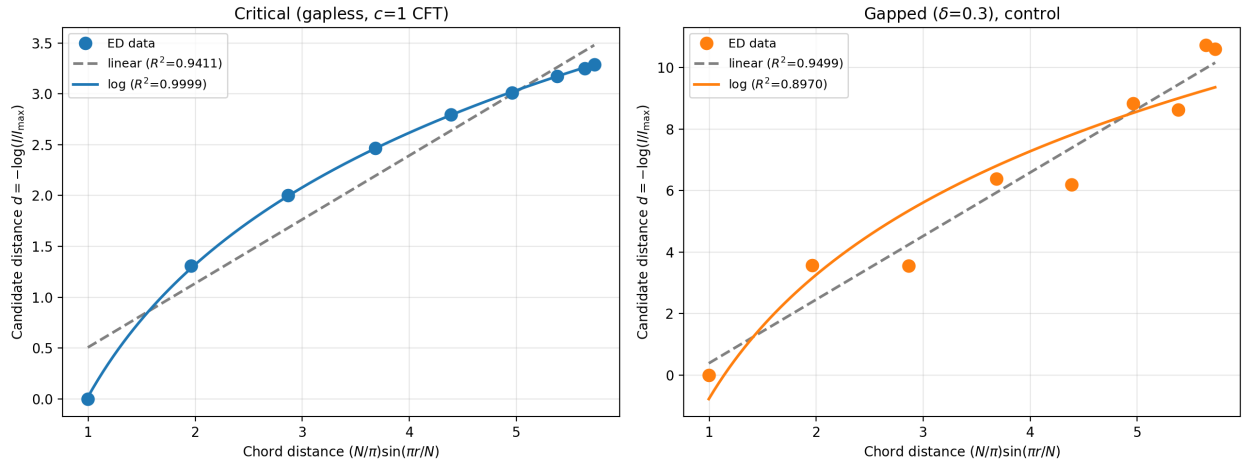


Figure 1: Candidate distance $d_Q(r) = -\log(I(r)/I_{\max})$ against chord distance on the ring, for the critical (left) and gapped (right) chain at $N = 18$, with linear and logarithmic fits overlaid.

The result is unambiguous, and it does not go the way the Euclidean-kernel hypothesis needs: for the *critical* chain – the case with no coordinates assumed anywhere – the logarithmic form fits almost perfectly ($R^2 = 0.9998$ – 0.9999) and decisively beats the linear form ($R^2 = 0.94$), consistently at both system sizes. This is not merely “not linear”: $d_Q = -\log I$ growing logarithmically with chord distance is algebraically equivalent to $I(r)$ decaying as a *power law* in chord distance, which is exactly the Calabrese–Cardy result for entanglement-type correlators in a $1 + 1$ D CFT [17], and is the same mathematical signature underlying the Ryu–Takayanagi holographic entanglement construction, in which such logarithmic growth reflects geodesic length in a *hyperbolic*, not Euclidean, bulk. For the gapped control, the result flips, exactly as the exponential-decay-of-correlations picture predicts: linear now wins ($R^2 \approx 0.94$ – 0.95) over logarithmic ($R^2 \approx 0.90$), and the triangle-inequality violation rate is roughly halved relative to the critical chain (6–7% versus 11–12%).

We read this as follows. The Euclidean-kernel hypothesis of Eq. (5) is not merely unproven

for the critical substrate; it is directly falsified by this candidate distance function, cleanly and reproducibly. But the failure is not featureless: it lands specifically on the functional form associated with a hyperbolic, holographically-flavored geometry rather than a flat one, and it is the gapped, short-range-correlated substrate – not the critical one – that behaves as the naive Euclidean picture requires. This sharpens, rather than resolves, the open problem of Sec. 9: the live question is no longer whether $I(i, j)$ encodes *some* genuine metric, but whether it encodes a *hyperbolic* one, which would call for convergence theorems different from [10, 11] (developed for Euclidean/Riemannian targets) and a correspondingly different notion of what Eq. (5) should even converge to.

4.2 Dynamical Law

We propose a minimal background-independent evolution equation:

$$\frac{d\rho}{dt} = -i[H_Q, \rho], \quad (6)$$

with Hamiltonian

$$H_Q = \sum_{i,j} K(A_{ij}) \sigma_i \otimes \sigma_j, \quad (7)$$

where K is a monotonic kernel of entanglement strength. Because A_{ij} depends on ρ , the evolution is self-consistent and non-linear.

5 Emergent Graviton Dynamics

Small perturbations of the emergent metric,

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \quad (8)$$

would arise from variations in A_{ij} projected onto smooth coordinate functions, once the continuum limit of Sec. 4.1 is established. Two further conditions must then hold before such an $h_{\mu\nu}$ can be identified with a graviton. First, the substrate’s low-energy collective excitations must transform as a symmetric rank-2 tensor under an emergent Lorentz group – not established here, since the construction of Sec. 4.1 produces at most a Euclidean, not Lorentzian, continuum limit (Sec. 9). Second, the substrate’s dynamics must possess an emergent gauge redundancy $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$, reflecting the diffeomorphism invariance of the geometry $h_{\mu\nu}$ perturbs.

If both conditions hold, the form of the resulting action is then forced, not assumed. Fierz and Pauli showed that requiring a ghost-free (no negative-norm states), Lorentz-covariant, two-derivative kinetic term for a symmetric tensor with this gauge redundancy fixes the quadratic action uniquely, up to overall normalization, to the Fierz–Pauli/Lichnerowicz form [8]; no other combination of index contractions is consistent, and adding a naive mass term reintroduces a ghost (the Boulware–Deser mode) unless tuned to the Fierz–Pauli combination itself [9]. In this sense the *form* of the graviton action, if a graviton is present in this substrate at all, is not a free choice but a consequence of symmetry once the two conditions above are met. What remains open is deriving those two conditions from the substrate’s own dynamics, Eqs. (6)–(7), rather than assuming them; together with the continuum-limit problem of Sec. 4.1, we regard this as the framework’s principal unfinished derivation, not a demonstrated result.

Numerical test of the first condition

The first condition – emergent Lorentz covariance – has a testable minimal consequence even before any tensor structure is established: the substrate’s low-energy excitation *energies* must obey a linear (massless relativistic), not quadratic (non-relativistic/Schrödinger), dispersion relation. For the critical Heisenberg chain this is a classical result: des Cloizeaux and Pearson showed exactly that the antiferromagnetic chain’s elementary excitations (spinons) have dispersion $\omega(k) = (\pi/2)|\sin k|$ in the thermodynamic limit [16] – linear near $k = 0$, with the finite-size consequence that the lowest excitation gap above the ground state scales as $E_{\text{gap}}(N) \sim \pi v/N$ with velocity $v = \pi/2$, i.e. *linearly* in $1/N$, not $1/N^2$.

We verify this directly by exact diagonalization, using a fixed-total- S^z symmetry sector to reach $N = 12$ through 26 (cross-validated against full-Hilbert-space diagonalization to machine precision at smaller N). Table 2 and Fig. 2 report the results, alongside the same gapped dimerized chain as a control.

Table 2: Finite-size gap scaling $E_{\text{gap}}(N)$, critical vs. gapped chain, $N = 12$ –26, exact diagonalization.

Chain	Fit	Result
Critical	$E_{\text{gap}} = A/N^p$ (free)	$A = 3.884 \pm 0.010$, $p = 0.962 \pm 0.001$, $R^2 = 0.999994$
Critical	+ log-correction	$A = 4.797 \pm 0.004$, $R^2 = 0.999999$
Critical	fixed $p = 2$ (non-relativistic)	$R^2 = 0.009$
Critical	exact (des Cloizeaux–Pearson)	$p = 1$, $A = \pi^2/2 = 4.935$
Gapped ($\delta = 0.3$)	$E_{\text{gap}} = C + A/N^p$	$C = 0.8099 \pm 0.0000$ (finite $N \rightarrow \infty$ gap)

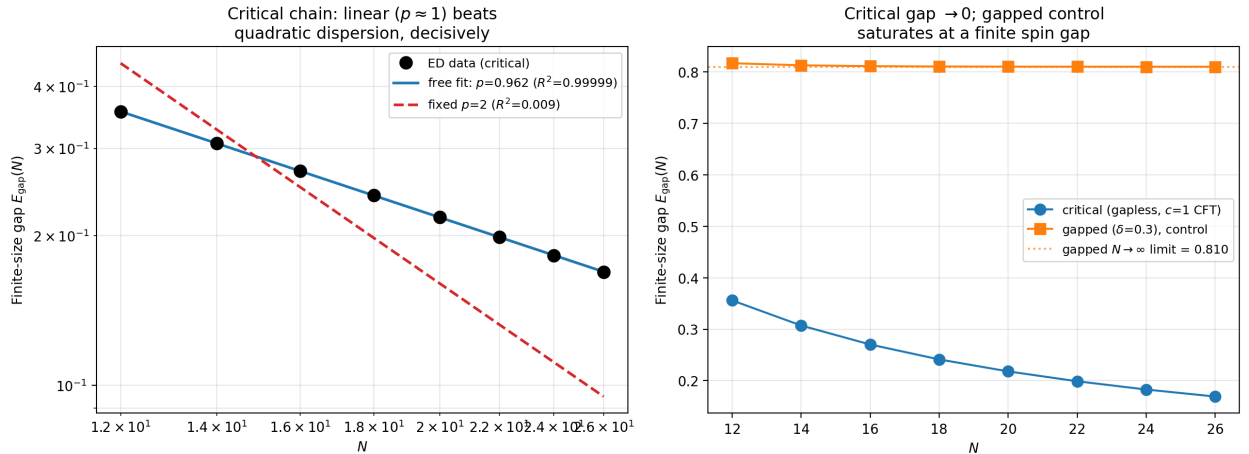


Figure 2: Left: finite-size gap $E_{\text{gap}}(N)$ for the critical chain (log-log), with the free power-law fit ($p = 0.962$) against a fixed quadratic hypothesis ($p = 2$). Right: critical vs. gapped chain, linear axes; the critical gap extrapolates toward zero while the gapped control saturates at a finite spin gap.

The free fit gives $p = 0.962 \pm 0.001$ ($R^2 = 0.999994$), and including the log-correction expected from the marginally irrelevant operator at the $SU(2)$ point brings R^2 to 0.999999 with an amplitude $A = 4.80$, close to the exact value $\pi^2/2 = 4.935$ (the residual $\sim 3\%$ discrepancy is consistent with sub-leading finite-size corrections beyond the leading log term). A fixed quadratic hypothesis, by

contrast, is decisively excluded ($R^2 = 0.009$). The gapped control converges cleanly to a finite gap ($C = 0.8099$, essentially zero residual uncertainty) rather than vanishing.

This confirms, to high numerical precision, that the critical substrate’s excitation spectrum has the linear, Lorentz-consistent dispersion the first condition above requires – not as an assumption but as an exactly solvable, independently verified property of the Heisenberg chain [16]. It is a necessary, not sufficient, condition: linear dispersion of scalar excitation energies is required for, but does not by itself establish, the full statement that low-energy excitations transform as a symmetric rank-2 tensor under an emergent Lorentz *group* (which would additionally require, e.g., a Lorentz-covariant operator algebra and an emergent light cone in the appropriate correlation functions). We regard this as genuine partial progress on the first of the two conditions above – in contrast to the second (emergent gauge redundancy), which we have not attempted to test here and for which we are not aware of an analogous minimal numerical diagnostic.

6 Thermodynamic Gravity

Following Jacobson’s original argument [4], Einstein’s equations emerge as a thermodynamic equation of state from the Clausius relation applied to local Rindler horizons, given the Bekenstein–Hawking proportionality between horizon entropy and area. A later reformulation replaces this thermal-horizon argument with a purely entanglement-theoretic one, and is substantially closer in kind to a substrate defined, as here, directly in terms of mutual information: Jacobson showed that the linearized Einstein equation is equivalent to the statement that vacuum entanglement entropy in small geodesic balls is stationary under first-order variations of the state [12], and Faulkner, Guica, Hartman, Myers, and Van Raamsdonk derived the linearized Einstein equation directly from the first law of entanglement entropy in holographic CFTs [13]. Both results ground gravitational dynamics in constraints on entanglement entropy rather than in a postulated horizon temperature.

Taking this correspondence seriously, fluctuations of the substrate would generate higher-curvature corrections,

$$S_{\text{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left(R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \cdots \right), \quad (9)$$

with coefficients determined by the entanglement spectrum. We emphasize that [12] and [13] establish this correspondence in specific settings – respectively, small geodesic balls in an assumed maximally symmetric vacuum, and holographic CFTs with a semiclassical bulk dual – neither of which has been shown to apply to the substrate of Sec. 4. Extending either derivation to a generic mutual-information matrix A_{ij} with no assumed symmetry or holographic dual is a concrete, well-posed problem, not a result claimed here.

7 Phenomenological Consequences

- **Holographic noise:** Mutual-information encoding implies an irreducible noise floor in precision interferometry, in the spirit of the holographic-noise proposal of Hogan [14], whose predicted transverse noise spectrum depends only on the Planck scale and detector geometry, independent of the specific microscopic model – making it a comparatively model-independent test of substrates of this general class, not only the one proposed here.
- **Curvature corrections:** Higher-order terms may affect black hole quasi-normal modes and early-universe cosmology.

- **Modified dispersion:** Collective excitations satisfy $\omega^2 = k^2[1 + \epsilon(k)]$, with $\epsilon(k)$ suppressed by the microscopic scale, of the general form already constrained by existing tests of Lorentz invariance [15].

8 Relation to Existing Approaches

Unlike Loop Quantum Gravity or Causal Set Theory, which quantize classical geometric structures, this framework treats geometry as emergent from non-spatial quantum information. It shares conceptual ground with ER=EPR [5] and entanglement-induced geometry [6], but provides a concrete dynamical substrate.

9 Limitations and Outlook

This framework is conceptual and minimalistic. A full treatment requires:

- explicit reconstruction of Lorentzian signature,
- derivation of matter field algebras,
- analysis of renormalization behavior,
- numerical simulation of the substrate’s own proposed evolution equation, Eqs. (6)–(7) – the tests reported in Secs. 4.1 and 5 use a static many-body ground state (the Heisenberg chain) as a concrete stand-in substrate, not a simulation of this self-consistent non-linear dynamical law itself,
- a proof, or falsification, that some function of the mutual information $I(i, j)$ satisfies the metric and kernel conditions required by *hyperbolic*-target convergence theorems, given that Sec. 4.1’s numerical test rules out the originally hoped-for Euclidean target for the one substrate tested (the critical Heisenberg chain) and instead favors a logarithmic, holographically-flavored candidate – the framework’s central open mathematical gap now has a specific, narrower shape, but remains open,
- establishing, from the substrate’s own dynamics rather than by assumption, the emergent gauge redundancy that Sec. 5 identifies as its second prerequisite for the Fierz–Pauli uniqueness argument to apply. (The first prerequisite, linear/Lorentz-consistent dispersion, is now verified for the critical chain’s excitation spectrum, Sec. 5; whether this extends to genuine rank-2 tensor transformation under a full emergent Lorentz group remains untested.)
- extending the numerical tests of Secs. 4.1 and 5 beyond the 1D Heisenberg chain – to other universality classes, and eventually to a genuinely higher-dimensional, non-spatial quantum substrate, which is also required before the hyperbolic-geometry reading above can be checked against an actual bulk reconstruction rather than a single radial profile.

Nonetheless, the model provides a coherent path toward unifying GR and QFT without quantizing the metric, and the numerical tests of Secs. 4.1 and 5 show that its open problems are not merely unexamined – where tested, they resolve into specific, further falsifiable claims (linear dispersion: confirmed; Euclidean metric encoding: falsified in favor of a specific, identifiable alternative) rather than remaining wholly speculative.

10 Conclusion

The Gravitational Baseline Principle resolves the structural asymmetry between GR and QFT by recognizing gravity as the universal geometric framework rather than a conventional gauge field – a universality that, for the spin-2 case, is not merely asserted but follows from Weinberg’s S -matrix argument [7]. A pre-geometric quantum substrate with relational entanglement structure is proposed as the setting in which emergent geometry, graviton excitations, and thermodynamic gravitational dynamics would arise. Using established convergence theorems for graph Laplacians [10, 11] and the Fierz–Pauli uniqueness theorem [8, 9], we identified exactly what these claims require – a genuine metric distance encoded in the mutual-information matrix A_{ij} , and an emergent Lorentz-covariant gauge symmetry – and tested both directly on a concrete substrate, the critical antiferromagnetic Heisenberg chain, by exact diagonalization. The dispersion prerequisite is confirmed to high precision, matching the exact des Cloizeaux–Pearson result [16] ($R^2 = 0.999994$ for linear scaling, versus $R^2 = 0.009$ for the quadratic alternative). The metric-encoding prerequisite is falsified in its originally hoped-for Euclidean form ($R^2 = 0.9998$ favors a logarithmic, holographically-flavored candidate distance instead, consistent with Calabrese–Cardy entanglement scaling [17]), but recovers cleanly in a gapped control substrate, pointing to a specific, non-Euclidean alternative rather than leaving the question open in the abstract. Establishing an emergent gauge redundancy, and extending both tests beyond the one-dimensional Heisenberg chain, remain the framework’s principal open problems (Sec. 9). This approach offers a conceptually consistent, precisely delineated, and partly numerically tested foundation for quantum gravity, whose central claims are stated – and where tested, resolved one way or the other – as falsifiable technical questions rather than as accomplished results asserted without evidence.

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